

Command Cheat Sheet

Wednesday, August 26, 2026 9:33 AM

Command	Description	Mode Required	Day Learned
?	Example: Router# ? — Displays available commands/options	User EXEC > or Privileged EXEC #	Day 4
command ?	Example: Router# show ? — Shows available options for a specific command	Any mode	Day 4
TAB	Example: Router# conf + TAB → configure — Auto-completes a partially typed command	Any mode	Day 4
enable	Example: Router> enable → Router# — Moves from User EXEC to Privileged EXEC	User EXEC >	Day 4
disable	Example: Router# disable → Router> — Moves from Privileged EXEC to User EXEC	Privileged EXEC #	Day 4
configure terminal / conf t	Example: Router# conf t → Router(config)# — Enters Global Configuration Mode	Privileged EXEC #	Day 4
exit	Example: Router(config)# exit → Router# — Moves back one CLI level	Any configuration mode	Day 4
end	Example: Router(config-if)# end → Router# — Immediately returns to Privileged EXEC Mode	Any configuration mode	Day 4
do	Example: Router(config)# do show ip interface brief — Runs a Privileged EXEC command while in configuration mode	Global Config or lower	Day 4
enable password [pw]	Example: Router(config)# enable password cisco — Sets a password for entering Privileged EXEC mode	Global Config	Day 4
enable secret [pw]	Example: Router(config)# enable secret cisco — Sets an encrypted password for Privileged EXEC; preferred over enable password	Global Config	Day 4
service password-encryption	Example: Router(config)# service password-encryption — Encrypts plaintext passwords using weak Type 7 encryption	Global Config	Day 4
no service password-encryption	Example: Router(config)# no service password-encryption — Stops future automatic encryption of plaintext passwords	Global Config	Day 4
show running-config	Example: Router# show running-config — Displays the current active configuration	Privileged EXEC #	Day 4
show startup-config	Example: Router# show startup-config — Displays the configuration saved in NVRAM for the next reboot	Privileged EXEC #	Day 4
write / write memory	Example: Router# write memory — Saves running-config to startup-config	Privileged EXEC #	Day 4
copy running-config startup-config	Example: Router# copy running-config startup-config — Copies the active configuration into startup configuration	Privileged EXEC #	Day 4
no [command]	Example: Router(config)# no shutdown — Removes or disables a configuration command	Appropriate configuration mode	Day 4
show arp	Example: Router# show arp — Shows the device's ARP table	Privileged EXEC #	Day 6
show mac address-table	Example: Switch# show mac address-table — Shows a switch's learned MAC address table	Privileged EXEC #	Day 6
clear mac address-table dynamic	Example: Switch# clear mac address-table dynamic — Clears dynamically learned MAC-table entries	Privileged EXEC #	Day 6
ping [ip]	Example: Router# ping 192.168.1.1 — Tests IP reachability using ICMP Echo messages	User EXEC > or Privileged EXEC #	Day 6
interface g0/0	Example: Router(config)# interface g0/0 → Router(config-if)# — Enters configuration mode for GigabitEthernet 0/0	Global Config	Day 8
ip address [ip] [mask]	Example: Router(config-if)# ip address 192.168.1.1 255.255.255.0 — Assigns an IPv4 address and subnet mask to an interface	Interface Config	Day 8
show ip interface brief	Example: Router# show ip interface brief — Lists interfaces, IP addresses, assignment method, status, and protocol	Privileged EXEC #	Day 8
show interfaces	Example: Router# show interfaces g0/0 — Shows detailed interface information, including Layer 1, Layer 2, and some Layer 3 details	Privileged EXEC #	Day 8
show interfaces description	Example: Router# show interfaces description — Shows each interface's description	Privileged EXEC #	Day 8
description [text]	Example: Router(config-if)# description WAN to ISP — Adds a label/note to an interface	Interface Config	Day 8

show interfaces status	Example: Switch# show interfaces status — Shows port status, VLAN, duplex, speed, and interface type	Privileged EXEC #	Day 8
no shutdown	Example: Router(config-if)# no shutdown — Administratively enables an interface	Interface Config	Day 9
shutdown	Example: Router(config-if)# shutdown — Administratively disables an interface	Interface Config	Day 9
interface range f0/5 - 12	Example: Switch(config)# interface range f0/5 - 12 — Enters configuration mode for FastEthernet 0/5 through 0/12 simultaneously	Global Config	Day 9
show ip route	Example: Router# show ip route — Displays the router's IPv4 routing table	Privileged EXEC #	Day 11
ip route [network] [mask] [next-hop]	Example: Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2 — Creates a static route to a network using a next-hop IP address	Global Config	Day 11
ip route [network] [mask] [exit-interface]	Example: Router(config)# ip route 192.168.2.0 255.255.255.0 g0/1 — Creates a static route using an exit interface	Global Config	Day 11
ip route 0.0.0.0 0.0.0.0 [next-hop]	Example: Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.2 — Configures an IPv4 default route	Global Config	Day 11
ip route 0.0.0.0 0.0.0.0 [exit-interface]	Example: Router(config)# ip route 0.0.0.0 0.0.0.0 g0/1 — Configures a default route using an exit interface	Global Config	Day 11
ip route [network] [mask] [next-hop] [exit-interface]	Example: Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2 g0/1 — Configures a static route using both a next-hop IP and exit interface	Global Config	Day 11
no ip route [network] [mask] ...	Example: Router(config)# no ip route 192.168.2.0 255.255.255.0 10.0.0.2 — Removes a configured static route	Global Config	Day 11
show ip route static	Example: Router# show ip route static — Displays only static routes in the routing table	Privileged EXEC #	Day 11
show ip route connected	Example: Router# show ip route connected — Displays directly connected routes	Privileged EXEC #	Day 11
show ip route 0.0.0.0	Example: Router# show ip route 0.0.0.0 — Checks routing information for the default route	Privileged EXEC #	Day 16
show vlan brief	Example: Switch# show vlan brief — Displays VLANs configured on the switch and shows which access ports belong to each VLAN	Privileged EXEC #	Day 16
vlan [vlan-id]	Example: Switch(config)# vlan 10 — Creates VLAN 10 if it does not already exist and enters VLAN configuration mode	Global Config	Day 16
name [name]	Example: Switch(config-vlan)# name SALES — Assigns a descriptive name to the VLAN	VLAN Config (config-vlan)	Day 16
interface range [interfaces]	Example: Switch(config)# interface range f0/1 - 10 — Selects multiple interfaces so they can be configured at the same time	Global Config	Day 16
switchport mode access	Example: Switch(config-if)# switchport mode access — Forces the switch interface to operate as an access port	Interface Config	Day 16
switchport access vlan [vlan-id]	Example: Switch(config-if)# switchport access vlan 10 — Assigns an access port to VLAN 10	Interface Config	Day 16
switchport trunk encapsulation dot1q	Example: Switch(config-if)# switchport trunk encapsulation dot1q — Sets the trunk encapsulation protocol to IEEE 802.1Q on switches that support multiple trunk encapsulation types	Interface Config	Day 17
switchport mode trunk	Example: Switch(config-if)# switchport mode trunk — Manually configures the interface as a trunk port	Interface Config	Day 17
show interfaces trunk	Example: Switch# show interfaces trunk — Displays interfaces currently operating as trunks, native VLANs, encapsulation, and allowed VLANs	Privileged EXEC #	Day 17
switchport trunk allowed vlan [list]	Example: Switch(config-if)# switchport trunk allowed vlan 10,20,30 — Restricts the trunk so only the specified VLANs are allowed across it	Interface Config	Day 17
switchport trunk allowed vlan add [list]	Example: Switch(config-if)# switchport trunk allowed vlan add 40 — Adds VLANs to the current allowed VLAN list without replacing the existing list	Interface Config	Day 17
switchport trunk allowed vlan remove [list]	Example: Switch(config-if)# switchport trunk allowed vlan remove 20 — Removes VLANs from the trunk's allowed list	Interface Config	Day 17
switchport trunk allowed vlan except [list]	Example: Switch(config-if)# switchport trunk allowed vlan except 30 — Allows all VLANs except the VLANs specified	Interface Config	Day 17
switchport trunk allowed vlan none	Example: Switch(config-if)# switchport trunk allowed vlan none — Removes all VLANs from the trunk's allowed list	Interface Config	Day 17
switchport trunk allowed vlan all	Example: Switch(config-if)# switchport trunk allowed vlan all — Allows all VLANs across the trunk	Interface Config	Day 17
switchport trunk native vlan [vlan-id]	Example: Switch(config-if)# switchport trunk native vlan 99 — Changes the native VLAN used for untagged traffic on an 802.1Q trunk	Interface Config	Day 17
interface g0/0.[subinterface]	Example: Router(config)# interface g0/0.10 — Creates/enters a router subinterface, commonly used for router-on-a-stick inter-VLAN routing	Global Config	Day 17
encapsulation dot1q [vlan-id]	Example: Router(config-subif)# encapsulation dot1q 10 — Associates the router subinterface with VLAN 10 using 802.1Q tagging	Subinterface Config (config-	Day 17

encapsulation dot1q [vlan-id] native	Example: Router(config-subif)# encapsulation dot1q 99 native — Associates the subinterface with the trunk's native VLAN	Subinterface Config	Day 17
ip address [ip] [mask]	Example: Router(config-subif)# ip address 192.168.10.1 255.255.255.0 — Assigns the VLAN's default-gateway IP address to the router subinterface	Subinterface Config	Day 17
Layer 3 Switching Commands			
ip routing	Example: SW1(config)# ip routing — Enables Layer 3 IP routing on a multilayer switch, allowing it to route between networks/VLANs.	Global Config	Day 20
no switchport	Example: SW1(config-if)# no switchport — Converts a switch interface from a Layer 2 switchport into a Layer 3 routed port , allowing an IP address to be assigned directly to it.	Interface Config	Day 20
Spanning Tree Verification Commands			
show spanning-tree	Example: SW1# show spanning-tree — Displays STP information including root bridge, root ports, designated ports, costs, and port states.	Privileged EXEC	Day 20
show spanning-tree summary	Example: SW1# show spanning-tree summary — Displays a summarized overview of STP configuration and status.	Privileged EXEC	Day 20
show spanning-tree detail	Example: SW1# show spanning-tree detail — Displays detailed STP information, including topology changes and BPDU activity.	Privileged EXEC	Day 20
PortFast Commands			
spanning-tree portfast	Example: SW1(config-if)# spanning-tree portfast — Enables PortFast on an interface. The port can transition to forwarding immediately instead of waiting through normal STP convergence. Typically used on end-device/access ports .	Interface Config	Day 20
spanning-tree portfast default	Example: SW1(config)# spanning-tree portfast default — Enables PortFast by default on eligible access ports.	Global Config	Day 21
spanning-tree portfast disable	Example: SW1(config-if)# spanning-tree portfast disable — Explicitly disables PortFast on the interface.	Interface Config	Day 21
spanning-tree portfast trunk	Example: SW1(config-if)# spanning-tree portfast trunk — Enables PortFast on a trunk interface. Use only when the trunk connects to an appropriate endpoint, not another switch .	Interface Config	Day 21
BPDU Guard Commands			
spanning-tree bpduguard enable	Example: SW1(config-if)# spanning-tree bpduguard enable — Enables BPDU Guard directly on an interface. If a BPDU is received, the port is typically placed into an err-disabled state.	Interface Config	Day 21
spanning-tree portfast bpduguard default	Example: SW1(config)# spanning-tree portfast bpduguard default — Globally enables BPDU Guard on PortFast-enabled ports.	Global Config	Day 21
show errdisable recovery	Example: SW1# show errdisable recovery — Displays err-disable recovery settings and reasons/interfaces associated with err-disable conditions.	Privileged EXEC	Day 21
Rapid PVST+ Command			
spanning-tree mode rapid-pvst	Example: SW1(config)# spanning-tree mode rapid-pvst — Configures the switch to use Rapid PVST+ , Cisco's per-VLAN implementation of Rapid Spanning Tree.	Global Config	Day 21
Root Bridge & STP Path Selection			
spanning-tree vlan 1 root primary	Example: SW1(config)# spanning-tree vlan 1 root primary — Adjusts bridge priority to make the switch the preferred root bridge for VLAN 1.	Global Config	Day 21
spanning-tree vlan 1 root secondary	Example: SW2(config)# spanning-tree vlan 1 root secondary — Adjusts bridge priority so the switch is preferred as the backup root bridge.	Global Config	Day 21
spanning-tree vlan 1 cost [cost]	Example: SW1(config-if)# spanning-tree vlan 1 cost 10 — Manually changes an interface's STP cost for VLAN 1, influencing STP's path selection.	Interface Config	Day 21
spanning-tree vlan 1 port-priority [priority]	Example: SW1(config-if)# spanning-tree vlan 1 port-priority 64 — Changes STP port priority for VLAN 1, which can influence which port is selected when other criteria tie.	Interface Config	Day 21
BPDU Filter			
spanning-tree bpdupfilter enable	Example: SW1(config-if)# spanning-tree bpdupfilter enable — Enables BPDU filtering on an interface, preventing normal sending/processing of BPDUs. Use carefully because it can defeat STP's loop protection.	Interface Config	Day 21
Root Guard			
spanning-tree guard root	Example: SW1(config-if)# spanning-tree guard root — Enables Root Guard , preventing a neighboring switch from becoming the STP root through that interface.	Interface Config	Day 21
Loop Guard			
spanning-tree guard loop	Example: SW1(config-if)# spanning-tree guard loop — Enables Loop Guard on an interface. Helps prevent STP loops if expected BPDUs suddenly stop being received.	Interface Config	Day 21
spanning-tree loopguard default	Example: SW1(config)# spanning-tree loopguard default — Globally enables Loop Guard on applicable point-to-point interfaces.	Global Config	Day 21
Rapid PVST			
spanning-tree mode rapid-pvst	Example: SW1(config)# spanning-tree mode rapid-pvst — Enables Rapid PVST+. Cisco's	Global Config	Day 22

spanning-tree portfast		Example: SW1(config-if)# spanning-tree portfast — Enables PortFast on an interface, allowing an edge/access port to transition to forwarding immediately. Typically used on ports connected to end devices.	Interface Config	Day 22
spanning-tree link-type point-to-point		Example: SW1(config-if)# spanning-tree link-type point-to-point — Manually configures an RSTP interface as a point-to-point link, typically used for full-duplex switch-to-switch connections.	Interface Config	Day 22
spanning-tree link-type shared		Example: SW1(config-if)# spanning-tree link-type shared — Manually configures an RSTP interface as a shared link, typically associated with half-duplex/shared Ethernet.	Interface Config	Day 22
etherchannel - General				
show etherchannel load-balance		Example: SW1# show etherchannel load-balance — Displays the load-balancing method currently being used by EtherChannel.	Privileged EXEC	Day 23
show etherchannel summary		Example: SW1# show etherchannel summary — Displays a summary of EtherChannels, including port-channel numbers, protocols, member interfaces, and their states.	Privileged EXEC	Day 23
show etherchannel port-channel		Example: SW1# show etherchannel port-channel — Displays detailed information about configured EtherChannel port-channel interfaces.	Privileged EXEC	Day 23
show etherchannel load-balance				
port-channel load-balance src-dst-mac		Example: SW1(config)# port-channel load-balance src-dst-mac — Configures EtherChannel load balancing based on source and destination MAC addresses.	Global Config	Day 23
Layer 2 EtherChannel — Switch-to-Switch				
interface range [interfaces]		Example: SW1(config)# interface range g0/1 - 2 — Selects the physical interfaces that will become members of the EtherChannel.	Global Config	Day 23
channel-group 1 mode desirable		Example: SW1(config-if-range)# channel-group 1 mode desirable — Adds the selected interfaces to Port-Channel 1 using PAgP desirable mode .	Interface Config	Day 23
channel-protocol pagp		Example: SW1(config-if-range)# channel-protocol pagp — Explicitly selects PAgP. Usually unnecessary because desirable/auto already indicates PAgP.	Interface Config	Day 23
channel-protocol lacp		Example: SW1(config-if-range)# channel-protocol lacp — Explicitly selects LACP. Usually unnecessary because active/passive already indicates LACP.	Interface Config	Day 23
interface port-channel 1		Example: SW1(config)# interface port-channel 1 — Enters configuration mode for the logical EtherChannel interface.	Global Config	Day 23
switchport trunk encapsulation dot1q		Example: SW1(config-if)# switchport trunk encapsulation dot1q — Configures the Port-Channel to use 802.1Q trunk encapsulation. Only required on switches that support multiple encapsulation types.	Interface Config	Day 23
switchport mode trunk		Example: SW1(config-if)# switchport mode trunk — Makes the logical Port-Channel a Layer 2 trunk capable of carrying multiple VLANs.	Interface Config	Day 23
Layer 3 EtherChannel — Multilayer Switch / Routed Connection				
interface range [interfaces]		Example: SW1(config)# interface range g0/1 - 2 — Selects the physical interfaces that will become members of the Layer 3 EtherChannel.	Global Config	Day 23
no switchport		Example: SW1(config-if-range)# no switchport — Converts the selected physical interfaces into Layer 3 routed interfaces.	Interface Config	Day 23
channel-group 1 mode active		Example: SW1(config-if-range)# channel-group 1 mode active — Adds the interfaces to Port-Channel 1 using LACP .	Interface Config	Day 23
interface port-channel 1		Example: SW1(config)# interface port-channel 1 — Enters configuration mode for the logical Layer 3 Port-Channel.	Global Config	Day 23
no switchport		Example: SW1(config-if)# no switchport — Configures the Port-Channel as a Layer 3 routed interface.	Interface Config	Day 23
ip address [IP] [mask]		Example: SW1(config-if)# ip address 10.0.0.1 255.255.255.252 — Assigns an IPv4 address to the logical Layer 3 Port-Channel.	Interface Config	Day 23
OSPF				
router ospf [process-id]		Example: R1(config)# router ospf 1 — Enables OSPF and enters OSPF configuration mode. The process ID is locally significant and does not have to match neighboring routers.	Global Config	Day 26
network [IP] [wildcard-mask] area [area-id] doing 0.0.0.0 255.255.255.255 works or just IP like 10.10.4.2 0.0.0.0		Example: R1(config-router)# network 10.0.0.0 0.0.0.255 area 0 — Identifies router interfaces whose IP addresses match the network/wildcard combination and enables OSPF on those interfaces. Also places those interfaces into the specified OSPF area.	Router Config	Day 26
passive-interface [interface] passive-interface default (will make all interfaces passive)		Example: R1(config-router)# passive-interface g0/1 — Stops OSPF Hello messages from being sent out the interface, preventing neighbor relationships from forming there. The connected network can still be advertised by OSPF. Typically used on interfaces connected to end-user LANs rather than other OSPF routers.	Router Config	Day 26
default-information originate After running this setup a default route on that router		Example: R1(config-router)# default-information originate — Advertises a default route (0.0.0.0/0) into OSPF so neighboring OSPF routers can learn it. By default, the router must already have a default route in its routing table.	Router Config	Day 26

(i.g. if router 1 accesses the internet, setup 0.0.0.0 0.0.0.0 to the next hop			
router-id [router-id]	Example: R1(config-router)# router-id 1.1.1.1 — Manually configures the router's OSPF Router ID, a 32-bit value written like an IPv4 address that uniquely identifies the OSPF router.	Router Config	Day 26
maximum-paths [number]	Example: R1(config-router)# maximum-paths 4 — Configures the maximum number of equal-cost OSPF routes that can be installed in the routing table for load balancing.	Router Config	Day 26
distance [number]	Example: R1(config-router)# distance 100 — Changes the administrative distance of OSPF routes. OSPF's default administrative distance is 110 .	Router Config	Day 26
show ip protocols	Example: R1# show ip protocols — Displays information about the routing protocols currently running, including OSPF process ID, Router ID, advertised networks, passive interfaces, and administrative distance.	Privileged EXEC	Day 26
show ip ospf interface brief			
clear ip ospf process	Example: R1# clear ip ospf process — Restarts the OSPF process. Useful after changing certain OSPF settings such as the Router ID. This temporarily tears down OSPF neighbor relationships.	Privileged EXEC	Day 26

OSPF Cost & Interface Commands — Day 27

auto-cost reference-bandwidth [Mbps]	Example: R1(config-router)# auto-cost reference-bandwidth 10000 — Changes the reference bandwidth OSPF uses to automatically calculate interface cost. Cost = Reference Bandwidth ÷ Interface Bandwidth . Configure the same reference bandwidth on all OSPF routers.	Router Config	Day 27
ip ospf cost [number]	Example: R1(config-if)# ip ospf cost 10 — Manually sets the OSPF cost of an interface, overriding the automatically calculated cost. Preferred method when you specifically want to change OSPF cost.	Interface Config	Day 27
bandwidth [Kbps]	Example: R1(config-if)# bandwidth 100000 — Changes the interface's logical bandwidth value , which can affect OSPF's automatic cost calculation. It does not change the interface's actual physical speed.	Interface Config	Day 27
show ip ospf neighbor	Example: R1# show ip ospf neighbor — Displays OSPF neighbors, including Router ID, priority, neighbor state, dead timer, neighbor IP address, and local interface.	Privileged EXEC	Day 27
ip ospf [process-id] area [area-id]	Example: R1(config-if)# ip ospf 1 area 0 — Enables OSPF directly on an interface and places it in the specified area. This is an alternative to using the OSPF network command.	Interface Config	Day 27

OSPF DR/BDR & Network Commands — Day 28

ip ospf priority [number]	Example: R1(config-if)# ip ospf priority 200 — Changes an interface's OSPF priority for DR/BDR elections . Higher priority is preferred. 0 makes the router ineligible to become DR or BDR on that interface.	Interface Config	Day 28
show ip ospf database	Example: R1# show ip ospf database — Displays the OSPF Link-State Database (LSDB) and the LSAs known by the router.	Privileged EXEC	Day 28
ip ospf network [type]	Example: R1(config-if)# ip ospf network point-to-point — Changes the OSPF network type used on an interface, affecting neighbor behavior and whether DR/BDR elections occur.	Interface Config	Day 28

OSPF Authentication Commands — Day 28

ip ospf authentication	Example: R1(config-if)# ip ospf authentication — Enables simple/plain-text OSPF authentication on the interface. Neighboring routers must use compatible authentication settings.	Interface Config	Day 28
ip ospf authentication-key [password]	Example: R1(config-if)# ip ospf authentication-key cisco — Configures the password used for simple OSPF authentication on the interface.	Interface Config	Day 28

Serial Interface Commands — Day 28

clock rate [bps]	Example: R1(config-if)# clock rate 128000 — Configures the clock rate on the DCE side of a serial connection. The DCE device provides clocking to the link.	Serial Interface Config	Day 28
encapsulation ppp	Example: R1(config-if)# encapsulation ppp — Changes the serial interface's Layer 2 encapsulation to PPP . Cisco serial interfaces traditionally default to HDLC.	Serial Interface Config	Day 28
show controllers [interface]	Example: R1# show controllers serial 0/0/0 — Displays hardware/controller information about an interface. On serial links, it can be used to determine which side is DCE or DTE .	Privileged EXEC	Day 28

MTU Commands — Day 28

ip mtu [bytes]	Example: R1(config-if)# ip mtu 1400 — Changes the maximum size of an IP packet that can be sent without fragmentation on the interface. OSPF neighbors need compatible MTU settings to successfully form a full adjacency.	Interface Config	Day 28
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IPv6 Commands — Day 31–32

ipv6 unicast-routing	Example: R1(config)# ipv6 unicast-routing — Enables the router to forward IPv6 packets between interfaces . Required for IPv6 routing.	Global Config	Day 31
ipv6 address [address/prefix]	Example: R1(config-if)# ipv6 address 2001:db8:1::1/64 — Manually assigns an IPv6 address and prefix length to an interface.	Interface Config	Day 31

show ipv6 interface brief	Example: R1# show ipv6 interface brief — Displays a summary of interfaces, their status, and assigned IPv6 addresses.	Privileged EXEC	Day 31
ipv6 address [prefix] eui-64	Example: R1(config-if)# ipv6 address 2001:db8:1::/64 eui-64 — Automatically generates the interface ID portion of the IPv6 address using modified EUI-64 .	Interface Config	Day 32
Standard Numbered ACLs — Day 34			
access-list [number] permit [source] [wildcard]	Example: R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255 — Creates a numbered standard ACL entry permitting traffic based only on the source IPv4 address . Standard ACL numbers are commonly 1–99 or 1300–1999.	Global Config	Day 34
access-list [number] deny [source] [wildcard]	Example: R1(config)# access-list 1 deny 192.168.1.0 0.0.0.255 — Creates a standard ACL entry denying traffic from the specified source network.	Global Config	Day 34
access-list [number] permit any	Example: R1(config)# access-list 1 permit any — Permits traffic from any source IPv4 address . any is shorthand for 0.0.0.0 255.255.255.255.	Global Config	Day 34
access-list [number] remark [text]	Example: R1(config)# access-list 1 remark ALLOW-SALES-LAN — Adds a description/comment to an ACL to document its purpose.	Global Config	Day 34
show access-lists	Example: R1# show access-lists — Displays configured ACLs, their entries, sequence numbers, and match counters.	Privileged EXEC	Day 34
show ip access-lists	Example: R1# show ip access-lists — Displays configured IPv4 ACLs and their entries.	Privileged EXEC	Day 34
ip access-group [ACL] in	Example: R1(config-if)# ip access-group 1 in — Applies an IPv4 ACL to traffic entering an interface.	Interface Config	Day 34
ip access-group [ACL] out	Example: R1(config-if)# ip access-group 1 out — Applies an IPv4 ACL to traffic leaving an interface.	Interface Config	Day 34
Named Standard ACLs — Day 35			
ip access-list standard [name]	Example: R1(config)# ip access-list standard BLOCK-SALES — Creates a named standard IPv4 ACL and enters Standard Named ACL configuration mode.	Global Config	Day 35
[sequence] permit [source] [wildcard]	Example: R1(config-std-nacl)# 10 permit 192.168.1.0 0.0.0.255 — Creates a permit ACE with a specific sequence number .	Standard Named ACL Config	Day 35
[sequence] deny [source] [wildcard]	Example: R1(config-std-nacl)# 20 deny 192.168.2.0 0.0.0.255 — Creates a deny ACE with a specific sequence number.	Standard Named ACL Config	Day 35
no [sequence]	Example: R1(config-std-nacl)# no 10 — Removes the ACL entry with that sequence number without deleting the entire ACL.	Named ACL Config	Day 35
ip access-list resequence [ACL] [starting-sequence] [increment]	Example: R1(config)# ip access-list resequence BLOCK-SALES 10 10 — Renumbers ACL entries beginning at sequence 10 and increasing by 10.	Global Config	Day 35
Extended ACLs — Day 35			
access-list [number] permit [protocol] [source] [destination]	Example: R1(config)# access-list 100 permit ip 192.168.1.0 0.0.0.255 any — Creates a numbered extended ACL permitting traffic based on protocol, source, and destination.	Global Config	Day 35
access-list [number] deny [protocol] [source] [destination]	Example: R1(config)# access-list 100 deny tcp 192.168.1.0 0.0.0.255 any — Creates an extended ACL denying matching traffic based on protocol, source, and destination.	Global Config	Day 35
ip access-list extended [name]	Example: R1(config)# ip access-list extended WEB-FILTER — Creates a named extended IPv4 ACL and enters Extended Named ACL configuration mode.	Global Config	Day 35
eq [port]	Example: R1(config-ext-nacl)# permit tcp any any eq 443 — Matches traffic with a port equal to the specified value.	Extended Named ACL Config	Day 35
gt [port]	Example: R1(config-ext-nacl)# permit tcp any any gt 1023 — Matches ports greater than the specified value.	Extended Named ACL Config	Day 35
lt [port]	Example: R1(config-ext-nacl)# permit tcp any any lt 1024 — Matches ports less than the specified value.	Extended Named ACL Config	Day 35
neq [port]	Example: R1(config-ext-nacl)# permit tcp any any neq 22 — Matches ports not equal to the specified value.	Extended Named ACL Config	Day 35
range [start] [end]	Example: R1(config-ext-nacl)# permit tcp any any range 20 21 — Matches a range of TCP or UDP port numbers.	Extended Named ACL Config	Day 35
CDP Verification Commands — Day 36			
show cdp	Example: SW1# show cdp — Displays global CDP information including timers, holdtime, and CDP version.	Privileged EXEC	Day 36
show cdp traffic	Example: SW1# show cdp traffic — Displays statistics for CDP packets sent and received.	Privileged EXEC	Day 36
show cdp interface	Example: SW1# show cdp interface — Displays CDP information and status for CDP-enabled interfaces.	Privileged EXEC	Day 36

show cdp neighbors	Example: SW1# show cdp neighbors — Displays directly connected Cisco neighbors including Device ID, local interface, capabilities, platform, and remote Port ID.	Privileged EXEC	Day 36
show cdp neighbors detail	Example: SW1# show cdp neighbors detail — Displays detailed information about CDP neighbors, including IP addresses, software/platform information, interfaces, and other advertised information.	Privileged EXEC	Day 36
show cdp entry [device-id]	Example: SW1# show cdp entry R1 — Displays detailed CDP information about a specific neighbor .	Privileged EXEC	Day 36

CDP Configuration Commands — Day 36

cdp run	Example: SW1(config)# cdp run — Enables CDP globally on the device.	Global Config	Day 36
no cdp run	Example: SW1(config)# no cdp run — Disables CDP globally on the device.	Global Config	Day 36
cdp enable	Example: SW1(config-if)# cdp enable — Enables CDP on a specific interface.	Interface Config	Day 36
no cdp enable	Example: SW1(config-if)# no cdp enable — Disables CDP on a specific interface.	Interface Config	Day 36
cdp timer [seconds]	Example: SW1(config)# cdp timer 30 — Changes how frequently the device sends CDP advertisements.	Global Config	Day 36
cdp holdtime [seconds]	Example: SW1(config)# cdp holdtime 90 — Changes how long received CDP information is retained before being discarded.	Global Config	Day 36
cdp advertise-v2	Example: SW1(config)# cdp advertise-v2 — Enables CDP Version 2 advertisements.	Global Config	Day 36

LLDP Verification Commands — Day 36

show lldp	Example: SW1# show lldp — Displays global LLDP information including timers and status.	Privileged EXEC	Day 36
show lldp traffic	Example: SW1# show lldp traffic — Displays statistics for LLDP frames sent and received.	Privileged EXEC	Day 36
show lldp interface	Example: SW1# show lldp interface — Displays LLDP transmit/receive status and information for interfaces.	Privileged EXEC	Day 36
show lldp neighbors	Example: SW1# show lldp neighbors — Displays directly connected LLDP neighbors including Device ID, local interface, capabilities, and remote Port ID.	Privileged EXEC	Day 36
show lldp neighbors detail	Example: SW1# show lldp neighbors detail — Displays detailed information learned about all LLDP neighbors.	Privileged EXEC	Day 36
show lldp entry [device-id]	Example: SW1# show lldp entry R1 — Displays detailed LLDP information about a specific neighbor.	Privileged EXEC	Day 36

LLDP Configuration Commands — Day 36

lldp run	Example: SW1(config)# lldp run — Enables LLDP globally on the device.	Global Config	Day 36
no lldp run	Example: SW1(config)# no lldp run — Disables LLDP globally.	Global Config	Day 36
lldp transmit	Example: SW1(config-if)# lldp transmit — Enables transmission of LLDP advertisements on the interface.	Interface Config	Day 36
lldp receive	Example: SW1(config-if)# lldp receive — Enables receiving LLDP advertisements on the interface.	Interface Config	Day 36
lldp timer [seconds]	Example: SW1(config)# lldp timer 30 — Configures how frequently LLDP advertisements are sent.	Global Config	Day 36
lldp holdtime [seconds]	Example: SW1(config)# lldp holdtime 120 — Configures the LLDP holdtime advertised to neighboring devices.	Global Config	Day 36
lldp reinit [seconds]	Example: SW1(config)# lldp reinit 2 — Configures the delay before LLDP is reinitialized on an interface.	Global Config	Day 36

NTP Verification Commands

show clock	Example: R1# show clock — Displays the device's current software clock time and date.	Privileged EXEC	Day 37
show clock detail	Example: R1# show clock detail — Displays detailed software clock information, including the configured time source and timezone information.	Privileged EXEC	Day 37
show ntp status	Example: R1# show ntp status — Displays general NTP status, including whether the device is synchronized, its stratum, reference clock, and timing information.	Privileged EXEC	Day 37
show ntp associations	Example: R1# show ntp associations — Displays configured NTP peers/servers and their synchronization status.	Privileged EXEC	Day 37

Manual Clock & Calendar Commands

clock set [hh:mm:ss] [day] [month] [year]	Example: R1# clock set 12:30:00 14 September 2026 — Manually sets the device's software clock .	Privileged EXEC	Day 37
calendar set [hh:mm:ss] [day] [month] [year]	Example: R1# calendar set 12:30:00 14 September 2026 — Manually sets the device's hardware calendar on devices that support one.	Privileged EXEC	Day 37
clock update-calendar	Example: R1# clock update-calendar — Copies the current software clock → hardware calendar .	Privileged EXEC	Day 37
clock read-calendar	Example: R1# clock read-calendar — Copies the hardware calendar → software clock .	Privileged	Day 37

		EXEC	
Timezone & Daylight Saving Commands			
clock timezone [name] [hours-offset] [minutes-offset]	Example: R1(config)# clock timezone EST -5 — Configures the device's timezone and its offset from UTC.	Global Config	Day 37
clock summer-time [name] recurring [start] [end] [offset]	Example: R1(config)# clock summer-time EDT recurring — Configures recurring daylight saving/summer time. Additional arguments can specify exactly when DST starts and ends.	Global Config	Day 37
NTP Client Commands			
ntp server [ip-address]	Example: R2(config)# ntp server 10.0.0.1 — Configures an NTP server that this device will use to synchronize its clock.	Global Config	Day 37
ntp server [ip-address] prefer	Example: R2(config)# ntp server 10.0.0.1 prefer — Configures an NTP server and marks it as the preferred server when multiple NTP servers are available.	Global Config	Day 37
ntp source [interface]	Example: R2(config)# ntp source loopback0 — Configures the specified interface's address as the source address for NTP packets.	Global Config	Day 37
ntp update-calendar	Example: R2(config)# ntp update-calendar — Configures the device to periodically update its hardware calendar using the NTP-synchronized software clock.	Global Config	Day 37
NTP Server & Peer Commands			
ntp master [stratum]	Example: R1(config)# ntp master 5 — Configures the Cisco device to act as an NTP master/server. The optional value specifies the stratum advertised by the device.	Global Config	Day 37
ntp peer [ip-address]	Example: R1(config)# ntp peer 10.0.0.2 — Configures a symmetric NTP peer relationship with another device.	Global Config	Day 37
ntp peer [ip-address] key [number]	Example: R1(config)# ntp peer 10.0.0.2 key 1 — Configures an NTP peer and specifies the authentication key to use.	Global Config	Day 37
NTP Authentication Commands			
ntp authenticate	Example: R1(config)# ntp authenticate — Globally enables NTP authentication.	Global Config	Day 37
ntp authentication-key [number] md5 [key]	Example: R1(config)# ntp authentication-key 1 md5 CCNA — Creates an NTP authentication key using the specified key number and password.	Global Config	Day 37
ntp trusted-key [number]	Example: R1(config)# ntp trusted-key 1 — Marks the specified NTP authentication key as trusted.	Global Config	Day 37
ntp server [ip-address] key [number]	Example: R2(config)# ntp server 10.0.0.1 key 1 — Configures an NTP server and specifies the authentication key used when communicating with it.	Global Config	Day 37
DNS Verification Commands			
show hosts	Example: R1# show hosts — Displays configured hostname-to-IP mappings as well as hostnames learned and cached through DNS.	Privileged EXEC	Day 38
DNS Configuration Commands			
ip dns server	Example: R1(config)# ip dns server — Configures the Cisco device to act as a DNS server and answer DNS queries.	Global Config	Day 38
ip domain lookup	Example: R1(config)# ip domain lookup — Enables the device to perform DNS hostname lookups. DNS lookup is normally enabled by default.	Global Config	Day 38
no ip domain lookup	Example: R1(config)# no ip domain lookup — Disables DNS hostname lookups. Useful in labs to prevent IOS from trying to resolve mistyped commands as hostnames.	Global Config	Day 38
ip host [hostname] [IP]	Example: R1(config)# ip host SERVER1 192.168.1.10 — Creates a static hostname-to-IP address mapping in the local host table.	Global Config	Day 38
ip name-server [IP]	Example: R1(config)# ip name-server 8.8.8.8 — Configures the IP address of a DNS server the device can query to resolve hostnames.	Global Config	Day 38
ip domain name [domain]	Example: R1(config)# ip domain name example.com — Configures the device's default domain name. Also commonly used when configuring SSH.	Global Config	Day 38
DHCP Show Commands			
show ip dhcp binding	Example: R1# show ip dhcp binding — Displays DHCP clients that currently have IP addresses assigned by the router's DHCP server.	Privileged EXEC	Day 39
DHCP Global Config Commands			
ip dhcp excluded-address [low-address] [high-address]	Example: R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10 — Excludes a range of addresses from being assigned to DHCP clients.	Global Config	Day 39
ip dhcp pool [pool-name]	Example: R1(config)# ip dhcp pool LAN — Creates a DHCP pool and enters DHCP Pool Configuration mode. If the pool already exists, enters its configuration.	Global Config	Day 39
network [network] [subnet-mask]	Example: R1(dhcp-config)# network 192.168.1.0 255.255.255.0 — Specifies the subnet from which DHCP addresses will be assigned. Excluded addresses will not be given to clients.	DHCP Pool Config	Day 39
lease [days] [hours] [minutes]	Example: R1(dhcp-config)# lease 7 — Configures how long DHCP clients can keep their assigned addresses.	DHCP Pool Config	Day 39
lease infinite	Example: R1(dhcp-config)# lease infinite — Configures DHCP leases so they do not expire.	DHCP Pool Config	Day 39

default-router [IP]	Example: R1(dhcp-config)# default-router 192.168.1.1 — Specifies the default gateway provided to DHCP clients.	DHCP Pool Config	Day 39
dns-server [IP]	Example: R1(dhcp-config)# dns-server 8.8.8.8 — Specifies the DNS server address provided to DHCP clients.	DHCP Pool Config	Day 39
domain-name [domain]	Example: R1(dhcp-config)# domain-name example.com — Specifies the domain name provided to DHCP clients.	DHCP Pool Config	Day 39
option 43 ip [IP]	Example: R1(dhcp-config)# option 43 ip 10.0.0.10 — Configures DHCP Option 43. Cisco APs can use this option to learn the IP address of their Wireless LAN Controller (WLC) .	DHCP Pool Config	Day 39
DHCP Interface Commands			
ip helper-address [IP]	Example: R1(config-if)# ip helper-address 10.0.0.10 — Configures the interface to act as a DHCP relay , forwarding client DHCP broadcasts toward a DHCP server on another network. Configure it on the interface facing the DHCP clients.	Interface Config	Day 39
ip address dhcp	Example: R1(config-if)# ip address dhcp — Configures the router interface to act as a DHCP client and obtain its own IPv4 address automatically from a DHCP server.	Interface Config	Day 39
SNMP Configuration Commands			
snmp-server contact [contact-info]	Example: R1(config)# snmp-server contact Network-Admin — Configures contact information for the person responsible for the SNMP-managed device.	Global Config	Day 40
snmp-server location [location-info]	Example: R1(config)# snmp-server location Server-Room — Configures a description of the physical location of the SNMP-managed device.	Global Config	Day 40
snmp-server community [string] ro	Example: R1(config)# snmp-server community CCNA ro — Configures an SNMP community string with read-only (RO) access. The NMS can read information but cannot modify it.	Global Config	Day 40
snmp-server community [string] rw	Example: R1(config)# snmp-server community CCNA rw — Configures an SNMP community string with read-write (RW) access. The NMS can read and modify information.	Global Config	Day 40
snmp-server host [IP] version 2c [community-string]	Example: R1(config)# snmp-server host 192.168.1.100 version 2c CCNA — Specifies the NMS that will receive SNMP notifications, using SNMPv2c and the specified community string.	Global Config	Day 40
snmp-server enable traps [trap-types]	Example: R1(config)# snmp-server enable traps link — Enables specified SNMP trap/notification types to be sent to the configured NMS.	Global Config	Day 40
Syslog Privileged Commands			
terminal monitor	Example: R1# terminal monitor — Displays Syslog messages in the current VTY session (SSH/Telnet) . Must be enabled again when a new remote session is established.	Privileged EXEC	Day 41
Syslog Configuration Commands			
logging console [level]	Example: R1(config)# logging console warnings — Sets the minimum Syslog severity level displayed on the console connection .	Global Config	Day 41
logging monitor [level]	Example: R1(config)# logging monitor warnings — Sets the Syslog severity level that can be displayed on VTY (SSH/Telnet) sessions when terminal monitor is enabled.	Global Config	Day 41
logging buffered [size] [level]	Example: R1(config)# logging buffered 16384 warnings — Stores Syslog messages in the device's RAM logging buffer . Optional size is specified in bytes.	Global Config	Day 41
logging trap [level]	Example: R1(config)# logging trap warnings — Sets the Syslog severity level sent to configured external Syslog servers .	Global Config	Day 41
logging [IP]	Example: R1(config)# logging 192.168.1.100 — Configures an external Syslog server to receive logging messages.	Global Config	Day 41
logging synchronous	Example: R1(config-line)# logging synchronous — Prevents Syslog messages from disrupting commands you are typing by redisplaying the prompt/input on a new line.	Line Config	Day 41
service sequence-numbers	Example: R1(config)# service sequence-numbers — Adds sequence numbers to Syslog messages, making it easier to determine their order.	Global Config	Day 41
service timestamps log datetime	Example: R1(config)# service timestamps log datetime — Adds the current date and time to Syslog messages.	Global Config	Day 41
service timestamps log uptime	Example: R1(config)# service timestamps log uptime — Adds the amount of time since the device booted to Syslog messages.	Global Config	Day 41
SSH — Day 42			
hostname [name] ip domain name [domain]	Example: R1(config)# hostname R1 / R1(config)# ip domain name example.com — Configures the device hostname and domain name. Both are needed before generating the RSA keys used by SSH.	Global Config	Day 42
username [username] secret [password]	Example: R1(config)# username admin secret CCNA123 — Creates a local username and encrypted secret for authentication. Used with login local.	Global Config	Day 42
crypto key generate rsa	Example: R1(config)# crypto key generate rsa — Generates the RSA key pair used by SSH. Requires a hostname and domain name to be configured first.	Global Config	Day 42
ip ssh version [1 2]	Example: R1(config)# ip ssh version 2 — Configures the SSH version. SSHv2 is preferred because SSHv1 is obsolete and insecure.	Global Config	Day 42
line vty [line] line vty [start] [end]	Example: R1(config)# line vty 0 15 — Enters VTY line configuration mode. VTY lines are used for remote CLI connections such as SSH and Telnet.	Global Config	Day 42
login login local	Example: R1(config-line)# login local — login authenticates using the password configured	Line Config	Day 42

login login local	Example: R1(config-line)# login local — login authenticates using the password configured directly on the VTY line. login local authenticates using the device's local username/password database and is commonly used for SSH.	Line Config	Day 42
transport input [protocols]	Examples: R1(config-line)# transport input ssh / transport input ssh telnet / transport input none — Controls which remote-access protocols are permitted on the VTY lines. transport input ssh is preferred because it prevents insecure Telnet access.	Line Config	Day 42
exec-timeout [minutes] [seconds]	Example: R1(config-line)# exec-timeout 5 0 — Automatically disconnects the remote session after the specified period of inactivity.	Line Config	Day 42
access-class [ACL] in	Example: R1(config-line)# access-class 10 in — Applies an ACL to incoming VTY connections to control which source IP addresses are allowed to remotely access the device.	Line Config	Day 42
ip default-gateway [IP]	Example: SW1(config)# ip default-gateway 192.168.1.1 — Configures a default gateway on a Layer 2 switch so the switch's management IP can communicate with SSH clients on other networks. Not normally needed on a router or multilayer switch performing IP routing.	Global Config	Day 42
show ip ssh show version	Example: R1# show ip ssh / R1# show version — show ip ssh verifies the SSH configuration, version, timeout, and retries. show version displays IOS/device information.	Privileged EXEC	Day 42

FTP / TFTP — Day 43

show file systems	Example: R1# show file systems — Displays available file systems and storage locations on the device. Useful first to identify locations such as flash:.	Privileged EXEC	Day 43
show flash	Example: R1# show flash — Displays the contents of Flash memory. Use this to verify the IOS image/file you want to copy or check available storage.	Privileged EXEC	Day 43
show version	Example: R1# show version — Displays the currently running IOS version and image along with device, uptime, memory, and boot information.	Privileged EXEC	Day 43
ip ftp username [username] ip ftp password [password]	Example: R1(config)# ip ftp username admin / R1(config)# ip ftp password CCNA123 — Configures the credentials the Cisco device will use when connecting to an FTP server . FTP only; TFTP does not use authentication.	Global Config	Day 43
copy [source] [destination]	Examples: R1# copy tftp: flash: / R1# copy ftp: flash: / R1# copy flash: tftp: / R1# copy flash: ftp: / R1# copy running-config tftp: / R1# copy tftp: running-config — Copies files or configurations between the Cisco device and another location. Source comes first, destination second.	Privileged EXEC	Day 43
delete [file-path]	Example: R1# delete flash:old-ios.bin — Deletes a specified file. Commonly used to remove an old IOS image or unwanted file from Flash.	Privileged EXEC	Day 43
boot system [file-path]	Example: R1(config)# boot system flash:cisco-ios.bin — Configures which IOS image the device should attempt to load during its next boot.	Global Config	Day 43

NAT — Day 44

ip nat inside ip nat outside	Example: R1(config-if)# ip nat inside / R1(config-if)# ip nat outside — Identifies which router interfaces connect to the inside/private network and which connect to the outside/public network .	Interface Config	Day 44
ip nat inside source static [inside-local] [inside-global]	Example: R1(config)# ip nat inside source static 192.168.1.10 203.0.113.10 — Configures Static NAT , creating a permanent one-to-one mapping between an inside local (private) address and an inside global (public) address.	Global Config	Day 44
show ip nat translations show ip nat statistics	Example: R1# show ip nat translations / R1# show ip nat statistics — translations displays the NAT translation table. statistics displays NAT activity, configured interfaces, pools, and translation statistics.	Privileged EXEC	Day 44
clear ip nat translation *	Example: R1# clear ip nat translation * — Clears all dynamic NAT/PAT translations from the NAT translation table.	Privileged EXEC	Day 44

NAT Part 2 — Day 45

ip nat inside ip nat outside	Example: R1(config-if)# ip nat inside / R1(config-if)# ip nat outside — Identifies interfaces as inside (private/internal network) or outside (public/external network) for NAT.	Interface Config	Day 45
ip nat inside source static [inside-local] [inside-global]	Example: R1(config)# ip nat inside source static 192.168.1.10 203.0.113.10 — Configures Static NAT , permanently mapping one inside local/private address to one inside global/public address.	Global Config	Day 45
access-list [ACL] permit [source] [wildcard]	Example: R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255 — Identifies the inside/local addresses that are eligible to be translated by Dynamic NAT or PAT .	Global Config	Day 45
ip nat pool [name] [start-IP] [end-IP] netmask [mask] ip nat pool [name] [start-IP] [end-IP] prefix-length [prefix]	Example: R1(config)# ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask 255.255.255.0 — Creates a pool of inside global/public addresses for Dynamic NAT or PAT.	Global Config	Day 45
ip nat inside source list [ACL] pool [pool] ip nat inside source list [ACL] pool [pool] overload	Example: R1(config)# ip nat inside source list 1 pool PUBLIC / R1(config)# ip nat inside source list 1 pool PUBLIC overload — Connects the ACL to the NAT pool. Without overload = Dynamic NAT . With overload = PAT , allowing multiple inside hosts to share addresses using port numbers.	Global Config	Day 45
ip nat inside source list [ACL] interface [interface] overload	Example: R1(config)# ip nat inside source list 1 interface g0/1 overload — Configures PAT using the outside interface's IP address . Multiple inside devices can share a single public IPv4 address using different port numbers.	Global Config	Day 45

PoE / Power Policing — Day 46

power inline police power inline police action err-	Example: SW1(config-if)# power inline police — Enables PoE power policing on the	Interface	Day 46
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disable power inline police action log	interface. Default/err-disable action places the port into err-disabled state if the powered device (PD) exceeds its allowed power. action log instead generates a Syslog message and restarts power to the device rather than err-disabling the port.	Config	
show power inline police [interface]	Example: SW1# show power inline police g0/1 — Displays PoE power-policing information for the specified interface, including power usage and policing status.	Privileged EXEC	Day 46
Voice VLAN / VoIP — Day 46			
switchport mode access switchport access vlan [data-vlan]	Example: SW1(config-if)# switchport mode access / SW1(config-if)# switchport access vlan 10 — Configures the switchport as an access port and assigns normal data traffic to the specified access VLAN.	Interface Config	Day 46
switchport voice vlan [voice-vlan]	Example: SW1(config-if)# switchport voice vlan 20 — Assigns traffic from an attached IP phone to the specified voice VLAN. Voice traffic is tagged with the voice VLAN ID while a connected PC can use the regular access/data VLAN.	Interface Config	Day 46
Port Security — Day 49			
switchport mode access switchport mode trunk	Example: SW1(config-if)# switchport mode access — Statically configures the switchport's mode. Port security requires the port to be statically configured as an access or trunk port rather than a dynamic port. Most CCNA examples use access ports .	Interface Config	Day 49
switchport port-security	Example: SW1(config-if)# switchport port-security — Enables port security on the interface.	Interface Config	Day 49
switchport port-security maximum [number]	Example: SW1(config-if)# switchport port-security maximum 2 — Sets the maximum number of secure MAC addresses allowed on the interface. Default = 1.	Interface Config	Day 49
switchport port-security mac-address [MAC] switchport port-security mac-address sticky switchport port-security mac-address sticky [MAC]	Example: SW1(config-if)# switchport port-security mac-address sticky — Controls which MAC addresses are considered secure. A MAC can be manually configured, or sticky can dynamically learn MAC addresses and add them to the running configuration as sticky secure MACs.	Interface Config	Day 49
switchport port-security violation shutdown switchport port-security violation restrict switchport port-security violation protect	Example: SW1(config-if)# switchport port-security violation restrict — Configures what happens when an unauthorized MAC causes a violation. Shutdown err-disables the port, restrict drops offending traffic while counting/logging violations, and protect drops offending traffic with less notification. Shutdown is the default.	Interface Config	Day 49
switchport port-security aging time [minutes] switchport port-security aging type absolute switchport port-security aging type inactivity switchport port-security aging static	Example: SW1(config-if)# switchport port-security aging time 60 / switchport port-security aging type inactivity — Configures secure MAC aging. absolute ages entries after the configured time regardless of activity; inactivity ages them after being inactive. aging static allows statically configured secure MAC addresses to age.	Interface Config	Day 49
errdisable recovery cause psecure-violation errdisable recovery cause dhcp-rate-limit errdisable recovery cause arp-inspection	Example: SW1(config)# errdisable recovery cause psecure-violation — Allows the switch to automatically recover ports that were err-disabled by the specified cause. These causes correspond to Port Security, DHCP Snooping, and Dynamic ARP Inspection .	Global Config	Day 49
errdisable recovery interval [seconds]	Example: SW1(config)# errdisable recovery interval 300 — Configures how long an interface remains err-disabled before the switch attempts automatic recovery for enabled recovery causes.	Global Config	Day 49
show port-security show port-security interface [interface]	Example: SW1# show port-security / SW1# show port-security interface f0/1 — Displays overall port-security information or detailed information for a specific interface, including status, violation mode, maximum MACs, secure MACs, and violation count.	Privileged EXEC	Day 49
show mac address-table secure	Example: SW1# show mac address-table secure — Displays secure MAC addresses learned or configured through port security.	Privileged EXEC	Day 49
show errdisable recovery	Example: SW1# show errdisable recovery — Displays configured err-disable recovery causes, timers, and interfaces currently waiting for recovery.	Privileged EXEC	Day 49
DHCP Snooping — Day 50			
ip dhcp snooping ip dhcp snooping vlan [vlan-id]	Example: SW1(config)# ip dhcp snooping / SW1(config)# ip dhcp snooping vlan 10 — Enables DHCP Snooping globally and then for the specified VLAN. Both are required.	Global Config	Day 50
no ip dhcp snooping information option	Example: SW1(config)# no ip dhcp snooping information option — Disables insertion of DHCP Option 82 information into DHCP messages.	Global Config	Day 50
ip dhcp snooping trust	Example: SW1(config-if)# ip dhcp snooping trust — Marks the interface as trusted , allowing DHCP server messages such as Offer and ACK through it. Typically configured toward the legitimate DHCP server/uplink. Ports are untrusted by default.	Interface Config	Day 50
ip dhcp snooping limit rate [pps]	Example: SW1(config-if)# ip dhcp snooping limit rate 15 — Limits the number of DHCP messages allowed per second on the interface. Exceeding the configured rate can cause the interface to become err-disabled . Commonly configured on untrusted/client-facing ports.	Interface Config	Day 50
errdisable recovery cause dhcp-rate-limit errdisable recovery interval [seconds]	Example: SW1(config)# errdisable recovery cause dhcp-rate-limit / SW1(config)# errdisable recovery interval 300 — Allows a port err-disabled because of a DHCP rate-limit violation to automatically recover after the configured interval.	Global Config	Day 50
show ip dhcp snooping binding	Example: SW1# show ip dhcp snooping binding — Displays the DHCP Snooping binding table , which maps learned client MAC addresses to IP addresses, VLANs, interfaces, and lease information.	Privileged EXEC	Day 50
Dynamic ARP Inspection (DAI) — Day 51			
ip arp inspection vlan [vlan-id]	Example: SW1(config)# ip arp inspection vlan 10 — Enables Dynamic ARP Inspection (DAI) for the specified VLAN. DAI normally uses the DHCP Snooping binding table to determine	Global Config	Day 50

	whether ARP messages are legitimate.		
ip arp inspection validate src-mac ip arp inspection validate dst-mac ip arp inspection validate ip ip arp inspection validate src-mac dst-mac ip	Example: SW1(config)# ip arp inspection validate src-mac dst-mac ip — Enables additional ARP validation. src-mac checks the Ethernet source MAC against the ARP sender MAC, dst-mac checks the Ethernet destination MAC against the ARP target MAC where applicable, and ip checks for invalid/unacceptable IP addresses in ARP packets. Multiple checks can be enabled in one command .	Global Config	Day 50
arp access-list [name] permit ip host [IP] mac host [MAC] deny ip host [IP] mac host [MAC]	Example: SW1(config)# arp access-list STATIC-HOSTS → SW1(config-arp-nacl)# permit ip host 192.168.1.10 mac host 0011.2233.4455 — Creates an ARP ACL and permits/denies specific IP-to-MAC mappings. Useful for hosts with static IP addresses that aren't represented in the DHCP Snooping binding table.	Global Config / ARP ACL Config	Day 50
ip arp inspection filter [arp-acl] vlan [vlan-id]	Example: SW1(config)# ip arp inspection filter STATIC-HOSTS vlan 10 — Applies an ARP ACL to DAI for the specified VLAN.	Global Config	Day 50
ip arp inspection trust	Example: SW1(config-if)# ip arp inspection trust — Marks an interface as trusted for DAI. ARP messages received on trusted interfaces bypass normal DAI inspection. Interfaces are untrusted by default . Typically used on appropriate infrastructure/uplink ports.	Interface Config	Day 50
ip arp inspection limit rate [pps] burst interval [seconds]	Example: SW1(config-if)# ip arp inspection limit rate 15 burst interval 1 — Limits the number of ARP packets allowed on the interface during the specified interval. Exceeding the limit can place the interface into err-disabled state.	Interface Config	Day 50
errdisable recovery cause arp-inspection errdisable recovery interval [seconds]	Example: SW1(config)# errdisable recovery cause arp-inspection / SW1(config)# errdisable recovery interval 300 — Enables automatic recovery for ports err-disabled by an ARP inspection rate violation and specifies how long the switch waits before attempting recovery.	Global Config	Day 50
show ip arp inspection show ip arp inspection interfaces	Example: SW1# show ip arp inspection / SW1# show ip arp inspection interfaces — Displays DAI configuration/statistics and the DAI status, trust state, and rate information for interfaces.	Privileged EXEC	Day 50